B/L/TX

Reg. No: 444

Final Assessment Test – November 2024

VIT Vellore Institute of Technology

Course: BMAT205L - Discrete Mathematics and Graph Theory

Class NBR(s): 2538 / 2559 / 2562 / 2563 / 2564 / 2565 / 2567 / 2569 / 2570 / 2571 / 2572 / 2573 / 2576 / 2579 / 2581 / 2583 / 2586

Slot: C2+TC2+TCC2

Time: Three Hours

Max. Marks: 100

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer ALL Questions

(10 X 10 = 100 Marks)

1. a) Construct the truth table for [10]

$$\neg(P \vee (Q \wedge R)) \leftrightarrow (P \vee Q) \wedge (P \rightarrow R)$$

b) Find the Principal Conjunctive Normal Form of

$(P \wedge Q) \vee (\neg P \wedge Q) \vee (Q \wedge R)$ without using the truth table.

2. Prove or disprove the validity of following argument: [10]

"Babies are illogical", "Nobody is despised who can manage crocodiles", "Illogical people are despised". Hence, babies cannot manage crocodiles.

3. a) Let $\{S,*\}, \{T,\Delta\}$, and $\{V,\oplus\}$ be semigroups and $g: S \rightarrow T$ and $h: T \rightarrow V$ be semigroup homomorphisms. Then prove that the composition function $h \circ g: S \rightarrow V$ is a semigroup homomorphism from $\{S,*\}$ to $\{V,\oplus\}$. [5]

b) If Z_6 is the set of equivalence classes generated by the equivalence relation "congruence modulo 6", prove that $\{Z_6,\times_6\}$ is a monoid where the operation $\times_6$ is defined as [5]

$$[i] \times_6 [j] = [(i \times j)(mod\ 6)], \forall [i], [j] \in Z_6.$$

4. Given the generator matrix $G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$ corresponding to the encoding function $e: B^3 \rightarrow B^6$, find the corresponding parity check matrix and use it to decode the following received words, and hence find the original message. [10]

(i) 1 1 0 1 0 1 (ii) 0 0 1 1 1 1 (iii) 1 1 0 0 0 1 (iv) 1 1 1 1 1 1

5. Solve the recurrence relation $a_n = 4a_{n-1} - 4a_{n-2} + 4^n; n \geq 2$ given that $a_0 = 2$ and $a_1 = 8$ using the method of generating functions. [10]

6.a) (i) If S_n is the set of all divisors of the positive integer n and D is the relation of the 'division' viz., "aDb" if and only if "a divides b", verify whether $\{S_{42}, D\}$ is a complemented lattice by finding the complements of all elements. [10]

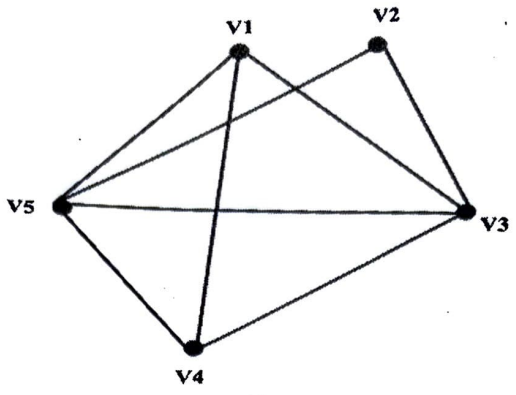
(ii) Simplify the Boolean expressions to a minimum number of literals $(xy'z' + xy'z + xyz + xyz')(x + y)$.

OR

6.b) Prove that every chain is a distributive Lattice. [10]

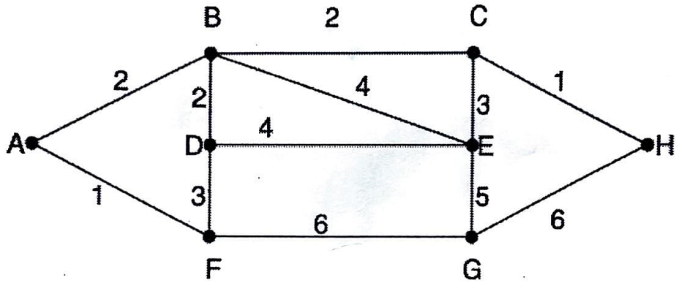
7. (i) Prove that the number of vertices of odd degrees in a graph is always even. **[10]**

(ii) Find the adjacency matrix of the following graph G.



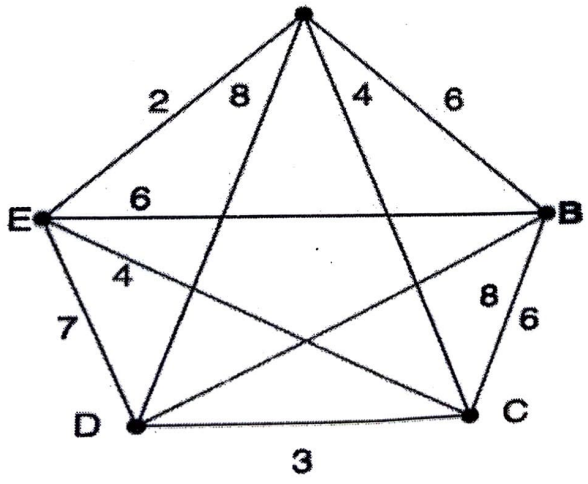
G

8. Use Dijkstra's algorithm to find the shortest path between the vertices A and H in the weighted graph given in G2 **[10]**



G2

9.a) Explain Krushkal's algorithm. Find the minimum spanning tree for the weighted graphs shown in G3 by using Krushkal's algorithm. **[10]**



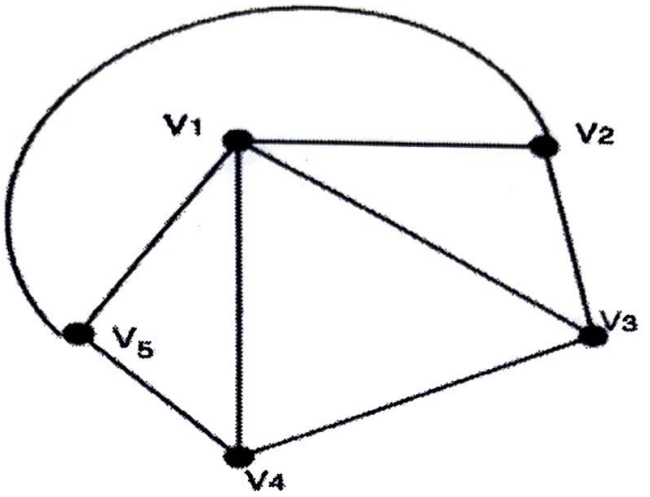
G3

OR

9.b) (i) Explain fundamental circuit with example. [5]

(ii) Prove that every tree has either one or two centers. [5]

10. Find the chromatic polynomial of the graph given in G4. [10]



G4

⇔⇔⇔B/L/TX⇔⇔⇔